

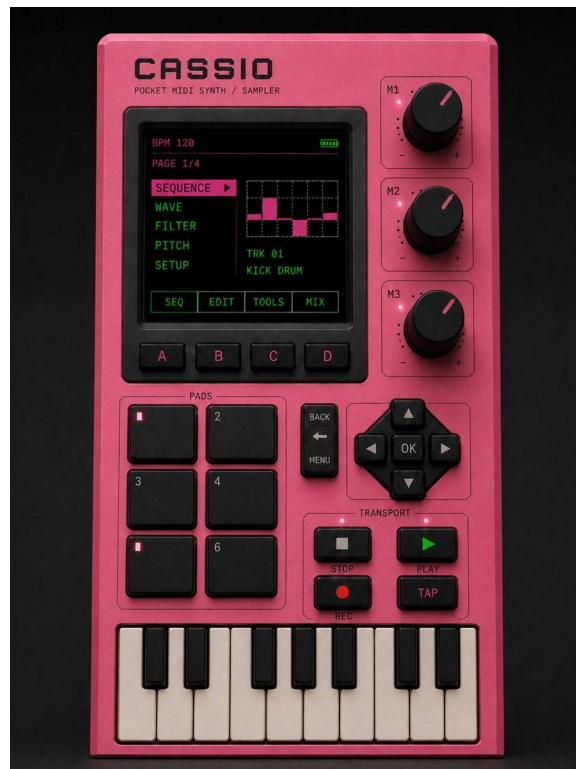
CASSIO

V1 COMPLETE OPERATING MANUAL

Pocket synth / sampler / looper - full front-panel behavior, every V1 display screen, onboard microphone sampling, file import, sound editing, sequencing, mixing, projects and settings.

V1 HARDWARE VERDICT

The current front panel is sufficient for the complete V1 feature set. No additional front-facing buttons or knobs are required. M1 and M2 are context controls, M3 is the predictable master-volume control in V1, A-D are screen-defined soft keys, the navigation cluster handles all menus, and the transport controls remain global.



CURRENT INDUSTRIAL-DESIGN REFERENCE

Production label correction: because V1 is a browser-based instrument and does not expose MIDI control, the enclosure subtitle should read "POCKET SYNTH / SAMPLER" rather than "POCKET MIDI SYNTH / SAMPLER."

What CASSIO V1 does

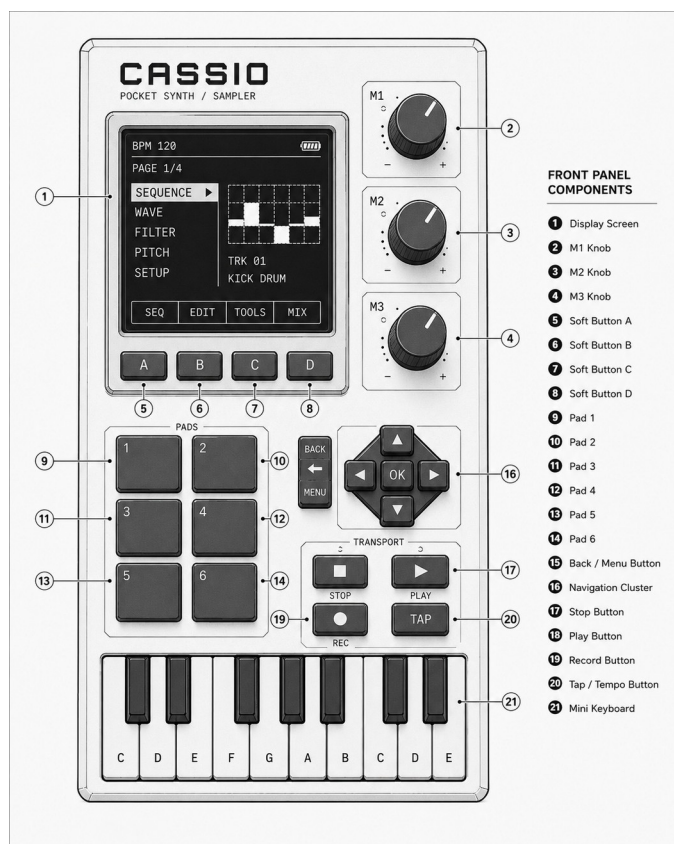
V1 is intentionally complete enough that the physical panel can be frozen. Later versions can add or rearrange digital screens without changing the enclosure.

AREA	V1 CAPABILITY
Performance	Chromatic mini keyboard, octave shifting, hold, factory and user sounds, six assignable pads.
Sound library	Factory library, user sounds, microphone recordings, imported files and favorites.
Sound shaping	Filter/shape, attack/release, two-band bass/treble EQ and simple reverb/delay.
Sampling	Record from the onboard microphone, import from the file system, trim, normalize, reverse, tune, set root note, select one-shot/gate/loop behavior, save and assign.
Looping	Six tracks, record, overdub, replace, mute, solo, undo, monitoring choices, loop length, count-in and quantization.
Sequencing	Six-lane step sequencer, 16-step base page, pattern length, swing, gate, step velocity/accent/tie and transport playback.
Mixing	Track level and pan, mute/solo, predictable master output level.
Projects	New, save, save-as, open, rename and automatic recovery state.
System	Master bass/treble EQ, metronome options, brightness, storage management and microphone permissions.

BASS + TREBLE

Yes. V1 should have a real two-band EQ. Per-sound/sample EQ uses M1 = Bass and M2 = Treble, each adjustable from -12 dB to +12 dB. Settings > Audio provides a second master Bass/Treble EQ after the mixer. In a Web Audio implementation these map cleanly to low-shelf and high-shelf filters.

Front panel - permanent control vocabulary



Display: All interface screens and contextual control labels.

M1: Context parameter 1.

M2: Context parameter 2.

M3: Master output volume in V1.

A-D: Four soft keys; meaning is printed at the bottom of each screen.

Pads 1-6: Performance sample pads; selected workflows may temporarily use them for lane selection.

BACK / MENU: Tap = back/cancel. Hold = main menu.

D-pad + OK: Move focus, edit values, enter, confirm.

STOP: Stop transport and return loop/pattern playhead to start. Hold = all-notes-off panic.

PLAY: Play/pause current loop or pattern.

REC: Start recording; during playback enter/leave overdub.

TAP: Tap tempo. Hold = metronome toggle.

Keyboard: Plays current keyboard sound.

Onboard mic: Always available as the sampler recording source; monitoring is off by default.

The current screen mockups assume M1/M2 are endless encoders if possible. That prevents jumps when software pages assign different parameters to the same knob. M3 may also be an encoder, but its V1 role remains master volume.

ENCLOSURE-LEVEL COMPONENTS

The onboard microphone, power control, charging connector and audio output are enclosure-level components rather than front-panel workflow controls. Their placement can be finalized mechanically without changing the button/knob vocabulary documented here.

Controls that behave the same everywhere

M1 / M2	Change the two parameters named across the top of the current display. Turning a knob opens a temporary value overlay.
M3	Always changes master output volume in V1. It never changes a saved track level or sample amplitude.
BACK / MENU	Tap BACK to leave the current screen, cancel a dialog or move up one level. Hold it to open the main menu from anywhere.
UP / DOWN	Move through lists or choose the previous/next editable row.
LEFT / RIGHT	Move horizontally, move a cursor, or decrease/increase a selected discrete value.
OK	Enter the highlighted item, toggle a step, confirm a choice, or open a selected row.
A-D	Execute the four actions printed at the bottom of the display. Blank labels do nothing.
PLAY	Start/pause loop or sequencer playback. While stopped, playback begins at the start.
STOP	Stop playback/recording and return to the beginning. Hold STOP for an immediate all-notes-off panic.
REC	From stopped: obey count-in then record the selected loop track. While playing: enter overdub. During recording: leave record but keep playback running.
TAP	Two or more taps set BPM from tap intervals. Hold TAP to toggle the metronome.
Keyboard	Remains playable on performance/edit screens unless a modal dialog is active.
Pads	Normally trigger assigned sounds. Screens that explicitly say LANE or PAD SELECT may repurpose them temporarily.

Status bar

Every normal screen shows BPM and battery state. Playback/recording state is indicated by a play or record symbol. No USB or MIDI status is included because those interfaces are not part of V1.

Screen map

Hold BACK / MENU to enter the six top-level areas. Everything else branches from them; no additional physical controls are required.

PLAY -> Library shortcut -> Sound Library -> Sound Detail -> Sound Edit / Pad Assign
LOOP -> Tracks -> OK Track Menu / C Options / B Sequencer -> Step Edit
SOUND -> Sound Library -> D Sampler -> Mic Record or File Import -> Sample Editor -> Save Sample
MIX -> six-track level/pan mixer
PROJECT -> project list -> Save / Save As / Rename / New
SETTINGS -> Audio / Metronome / Display / Storage / Permissions / About

Fast workflows

PLAY A SOUND	Boot -> PLAY -> D LIBRARY -> choose sound -> OK -> A USE KEYS.
RECORD MIC	Hold MENU -> SOUND -> D SAMPLER -> A MIC -> REC -> STOP -> edit -> save.
IMPORT SAMPLE	Hold MENU -> SOUND -> D SAMPLER -> B FILE -> D OPEN -> choose file -> edit -> save.
MAKE A LOOP	Hold MENU -> LOOP -> select track -> REC -> perform -> REC to leave overdub.
STEP PROGRAM	LOOP -> B SEQ -> pad selects lane -> arrows select step -> OK toggles it.
SAVE PROJECT	Hold MENU -> PROJECT -> B SAVE; use Text Entry if naming is required.



Boot / audio-engine initialization

The only startup screen. It confirms that the browser audio engine and local project store are initializing before performance controls become live.

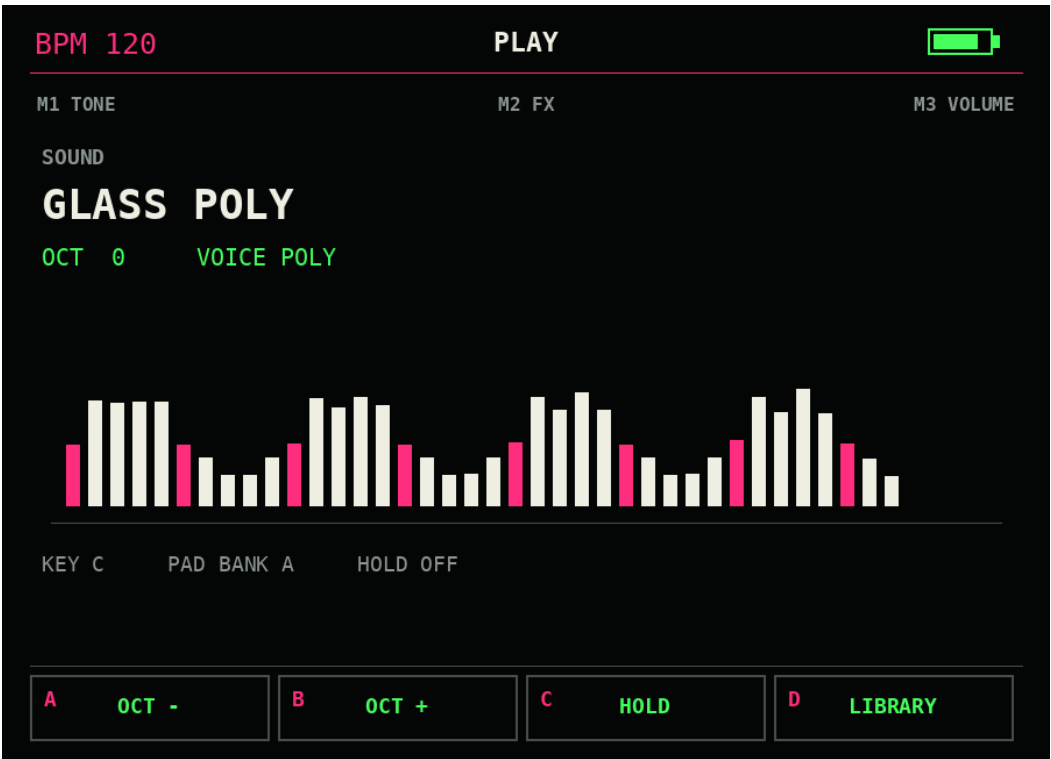


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Appears automatically after power/app launch.
HARDWARE	No performance input is accepted until loading completes. STOP may still be treated as a panic command if the engine is partially initialized.
WHAT IT DOES	No sound is produced until the audio context is ready. When initialization finishes, CASSIO enters PLAY using the last project or an empty project.
SAVE / STATE	Last-open project ID and recovery state are read from local persistent storage.
NOTES	If audio initialization fails, CASSIO should show an error instead of silently entering PLAY.

PLAY - performance home

The default performance screen. It keeps the selected sound, octave, voice state and two immediate sound macros visible while leaving the keyboard and pads playable.



HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Boot completes, or hold MENU -> PLAY.
HARDWARE	M1 = preset-defined Tone macro. M2 = preset-defined FX macro. M3 = master volume. A = octave down. B = octave up. C = Hold. D = open Library. Keyboard plays current sound; pads trigger assigned pad sounds.
WHAT IT DOES	Octave shifts keyboard notes by 12 semitones. Hold sustains released notes. M1/M2 alter the active sound in real time. Pad audio is independent of the keyboard sound.
SAVE / STATE	Selected keyboard sound, octave, Hold state and user edits are stored in the current project. Temporary macro movement becomes part of the sound state only when saving a user sound/project.
NOTES	Factory presets can be performed immediately. The current project key is metadata in V1 and does not force notes into a scale.

Main menu

The universal top-level navigator. It overlays the running instrument rather than stopping audio.

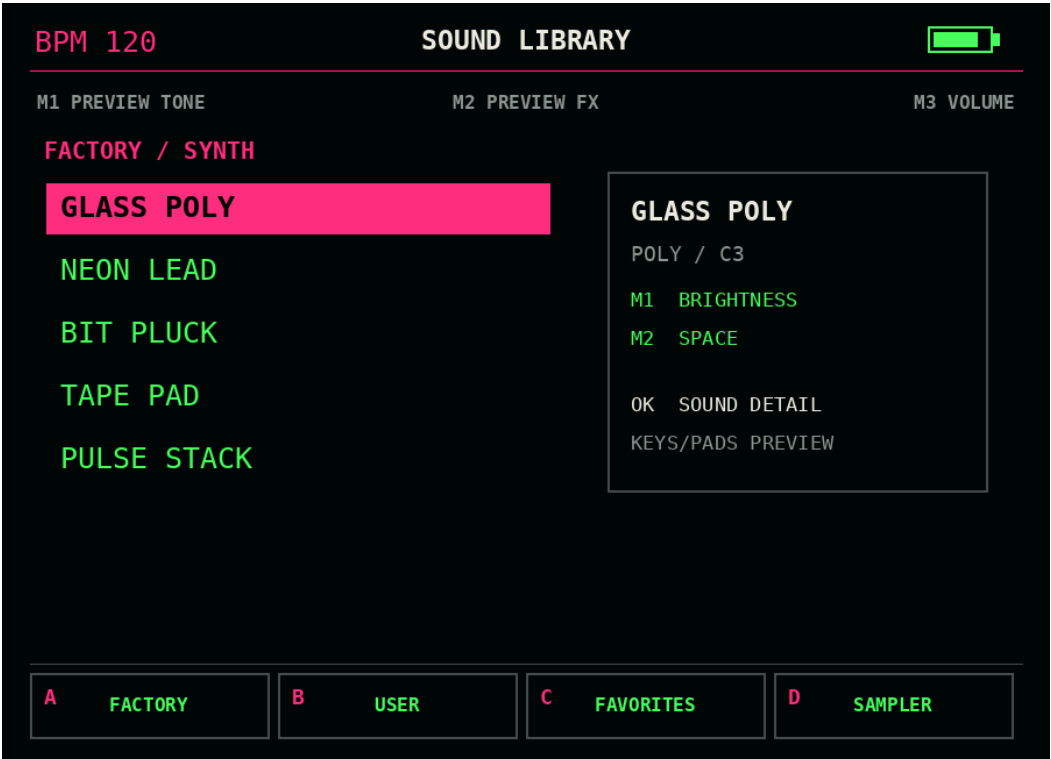


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Hold BACK / MENU from any normal screen.
HARDWARE	UP/DOWN moves through PLAY, LOOP, SOUND, MIX, PROJECT and SETTINGS. OK opens the highlighted area. BACK closes the menu. Transport stays active.
WHAT IT DOES	Opening the menu does not change the sound. Existing loop or sequencer playback continues unless STOP is pressed.
SAVE / STATE	No state is saved merely by opening or navigating the menu.
NOTES	This is the main future-proofing mechanism: new software screens can be added later without new hardware.

Sound Library

Browse all playable sounds: factory presets, user-created sounds, microphone recordings/imports, and favorites.

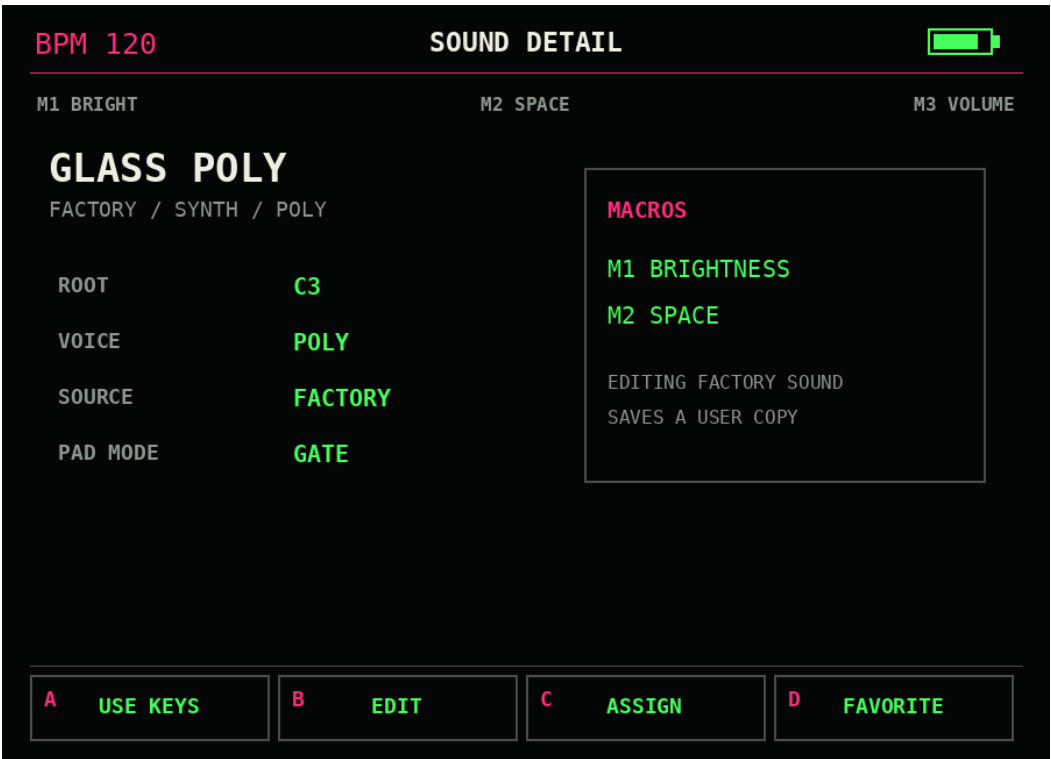


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	PLAY -> D LIBRARY, or hold MENU -> SOUND (the SOUND area opens the Library).
HARDWARE	A = Factory, B = User, C = Favorites, D = Sampler. LEFT/RIGHT changes category; UP/DOWN changes sound. Keyboard and pads audition the highlighted sound. OK opens Sound Detail; Sound Detail -> A USE KEYS confirms it. BACK exits without changing the loaded sound.
WHAT IT DOES	Highlighting temporarily previews the sound. USE KEYS commits it to the keyboard slot. M1/M2 preview the highlighted sound using that preset's macro assignments.
SAVE / STATE	Factory sounds are read-only. User contains edited sounds, microphone recordings and imported sounds. Favorites store references only.
NOTES	D SAMPLER is always the direct path from the Library to microphone recording or file import.

Sound Detail

Shows the selected sound's source, root note, voice mode and macro assignments before the user loads, edits or assigns it.

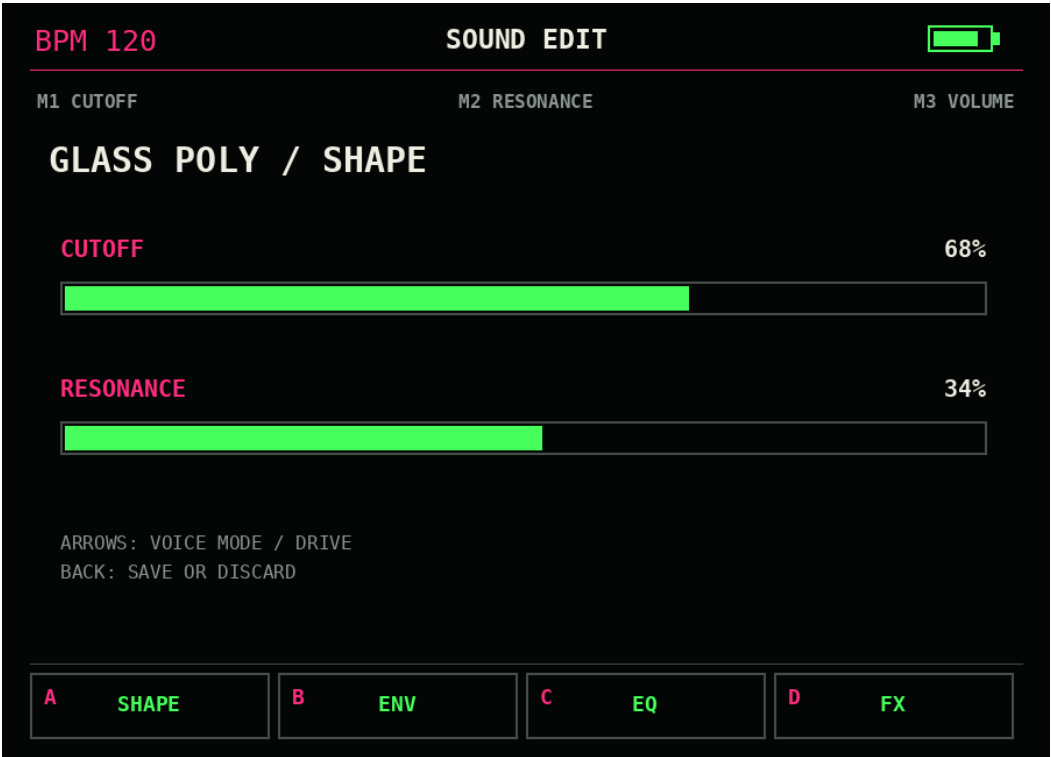


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	From Sound Library, highlight a sound and press OK/More, or open the current sound's detail action.
HARDWARE	A = use on keyboard. B = edit. C = assign to a pad. D = favorite/unfavorite. M1/M2 remain live so the sound can be auditioned.
WHAT IT DOES	Use Keys makes the sound the main pitched instrument. Assign maps it to one of the six pad slots. Edit opens non-destructive user parameters.
SAVE / STATE	Favorite status and pad/key assignment are project/library metadata. Editing a factory preset never overwrites factory content.
NOTES	Factory samples that are intended as one-shots may still be assigned to the keyboard; the root note defines pitch mapping.

Sound Edit - Shape

Primary tone shaping for a synth or saved user sound.

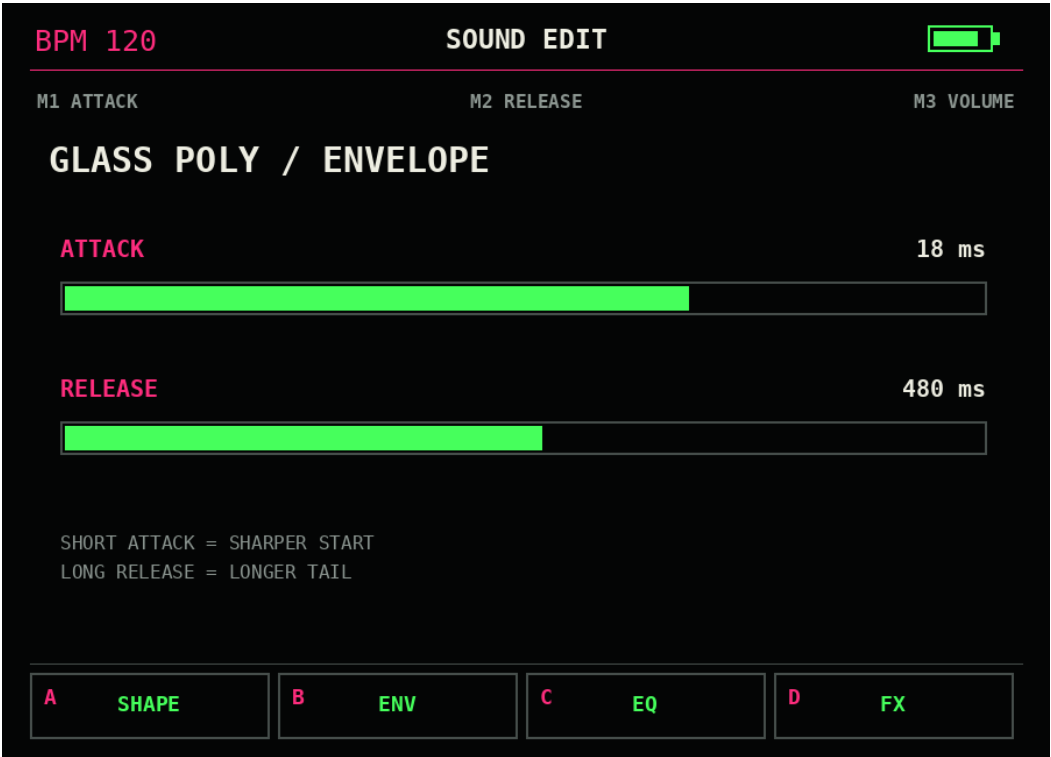


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Sound Detail -> B EDIT. A selects Shape page.
HARDWARE	M1 = cutoff/brightness. M2 = resonance. LEFT/RIGHT or OK can change discrete voice/drive options shown in the lower rows. A/B/C/D jump between Shape, Envelope, EQ and FX pages.
WHAT IT DOES	Cutoff removes or reveals high-frequency content. Resonance emphasizes frequencies around the cutoff. Voice/drive options can make a sound more focused, saturated or mono/poly depending on preset type.
SAVE / STATE	Edits remain pending until BACK; then Save Sound is shown. Factory sources require Save As.
NOTES	Parameter names can vary by factory preset, but V1 should prefer musically obvious controls rather than exposing raw synthesis internals.

Sound Edit - Envelope

Controls how quickly a sound begins and how long it lingers after release.

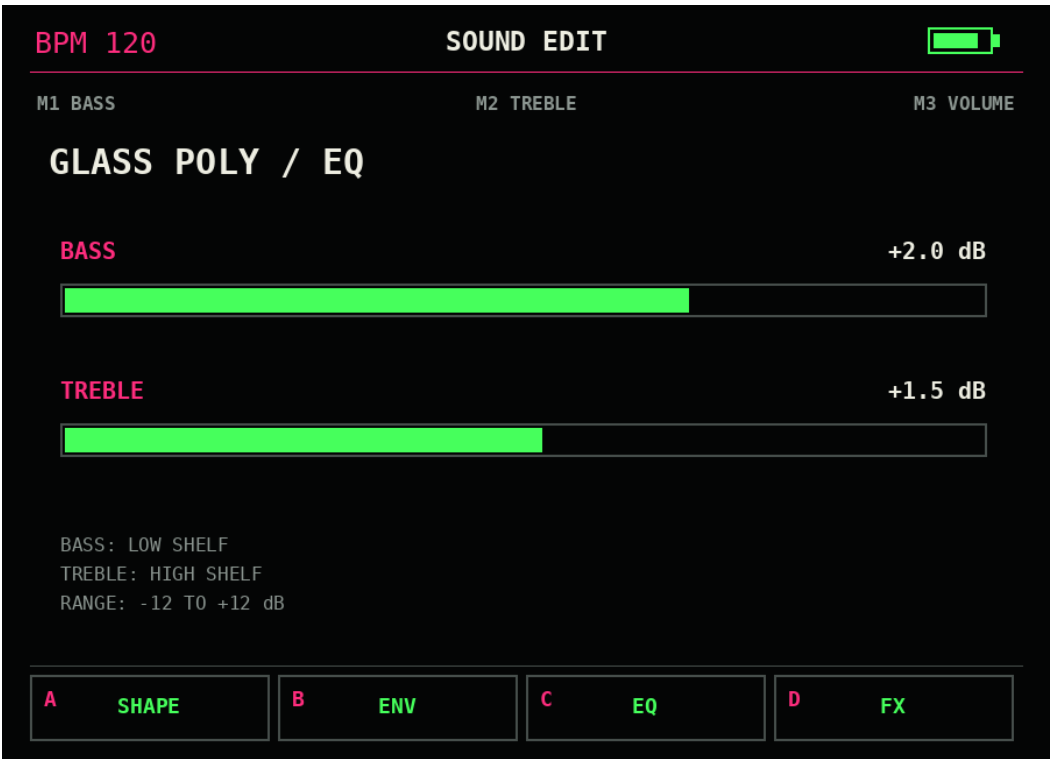


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Sound Edit -> B ENV.
HARDWARE	
WHAT IT DOES	M1 = Attack. M2 = Release. A/B/C/D switch edit pages.
SAVE / STATE	Short Attack produces a sharp immediate onset; longer Attack fades into the note. Short Release stops quickly; longer Release creates a tail after the key is released.
NOTES	Values are saved with a user sound or project sound state.
	V1 deliberately uses a simple two-control envelope here. More detailed ADSR controls can be added later as digital-only pages.

Sound Edit - EQ

Per-sound two-band tone correction.

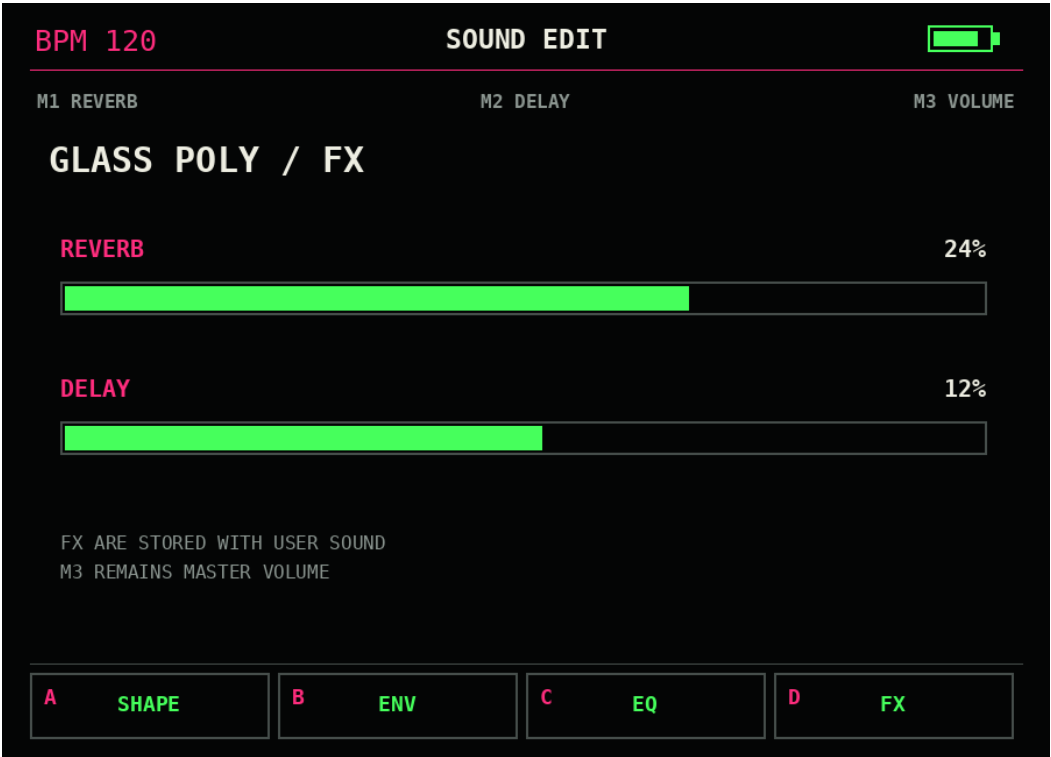


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Sound Edit -> C EQ.
HARDWARE	M1 = Bass, from -12 dB to +12 dB. M2 = Treble, from -12 dB to +12 dB. OK can reset the selected band to 0 dB; a Reset action may reset both.
WHAT IT DOES	Bass boosts/cuts low frequencies using a low-shelf EQ. Treble boosts/cuts high frequencies using a high-shelf EQ. This changes timbre, not note pitch or tempo.
SAVE / STATE	EQ values are stored with the user sound or current project sound state.
NOTES	Recommended implementation: low shelf around 120 Hz and high shelf around 6 kHz. Keep enough headroom/limiting to avoid clipping after boosts.

Sound Edit - FX

Simple send-style ambience and echo controls that are immediately playable.



HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE
HARDWARE
WHAT IT DOES
SAVE / STATE
NOTES

Sound Edit -> D FX.

M1 = Reverb amount. M2 = Delay amount. M3 still controls master volume.

Reverb adds room/space. Delay adds repeating echoes synchronized or free-running according to preset defaults. At 0% the corresponding effect is bypassed.

Values save with a user sound. Factory preset changes require Save As.

Detailed delay time, feedback or reverb type can be later digital pages; V1 only needs the two immediate musical amounts.

Save Sound

Common save dialog shown when leaving Sound Edit with changes.

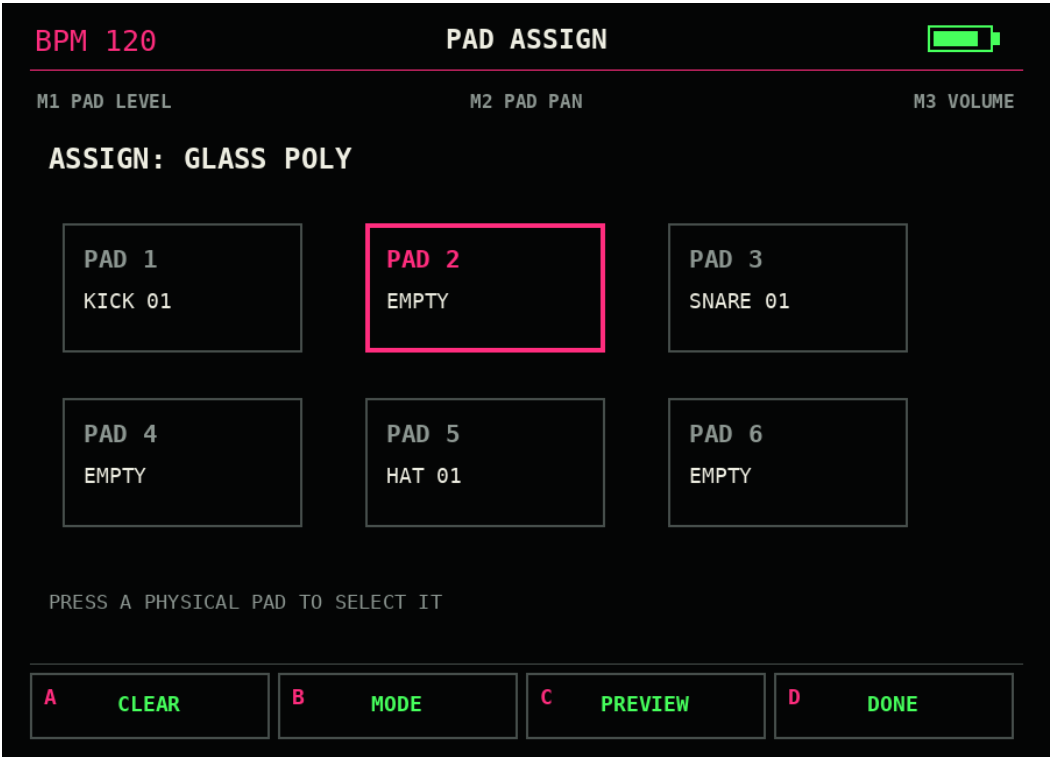


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Press BACK from Sound Edit after modifying a sound.
HARDWARE	A = cancel and return to editing. B = edit name. C = Save As. D = Save when editing an existing user sound. Factory sounds disable overwrite and require Save As.
WHAT IT DOES	Saving does not alter playback except that the edited sound becomes a reusable library item.
SAVE / STATE	User sound stores a reference to its source plus filter/envelope/EQ/FX settings; factory media stays untouched.
NOTES	This non-destructive model prevents accidental loss and keeps factory content immutable.

Pad Assignment

Maps a sound or sample to one of the six physical performance pads.

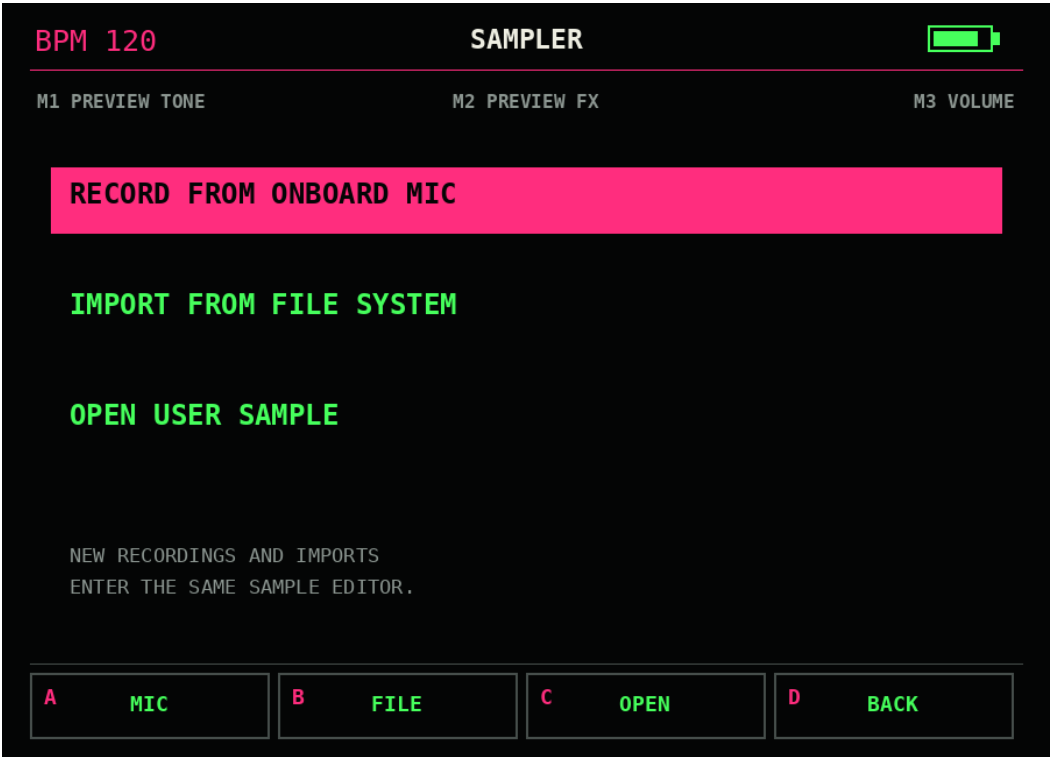


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Sound Detail -> C ASSIGN.
HARDWARE	Press a physical pad or use arrows to choose Pad 1-6. OK assigns the current sound. A clears the selected pad. B switches pad playback mode where allowed. C previews. D finishes. M1/M2 can set pad level/pan if shown.
WHAT IT DOES	Assigned pads become independent one-shot/gate/loop triggers during performance. Clearing removes the mapping but never deletes the sound from the library.
SAVE / STATE	Pad assignments save in the project. Optional per-pad level/pan save with the assignment.
NOTES	Pads remain performance triggers on PLAY and LOOP; the sequencer explicitly repurposes them as lane selectors.

Sampler Home

Entry point for creating a new user sound from the onboard microphone or an audio file.

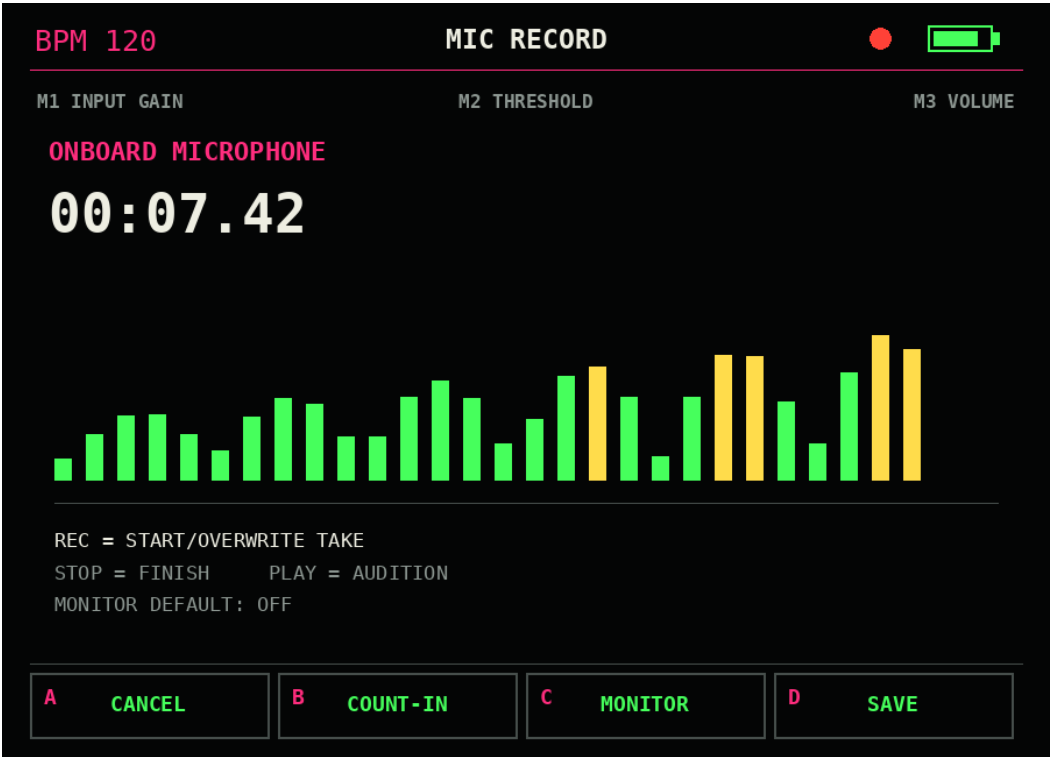


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Hold MENU -> SOUND to open the Sound Library, then press D = SAMPLER.
HARDWARE	A = microphone. B = file import. C = open an existing user sample. D = back. UP/DOWN and OK can also use the list.
WHAT IT DOES	No audio changes until a source is recorded/imported. Existing project playback may continue behind the sampler unless the user stops it.
SAVE / STATE	New recordings/imports are staged until the user reaches Save Sample.
NOTES	Microphone and imported sources enter the same editor, so the manipulation workflow is consistent.

Microphone Record

Records a new sample directly from the onboard microphone.

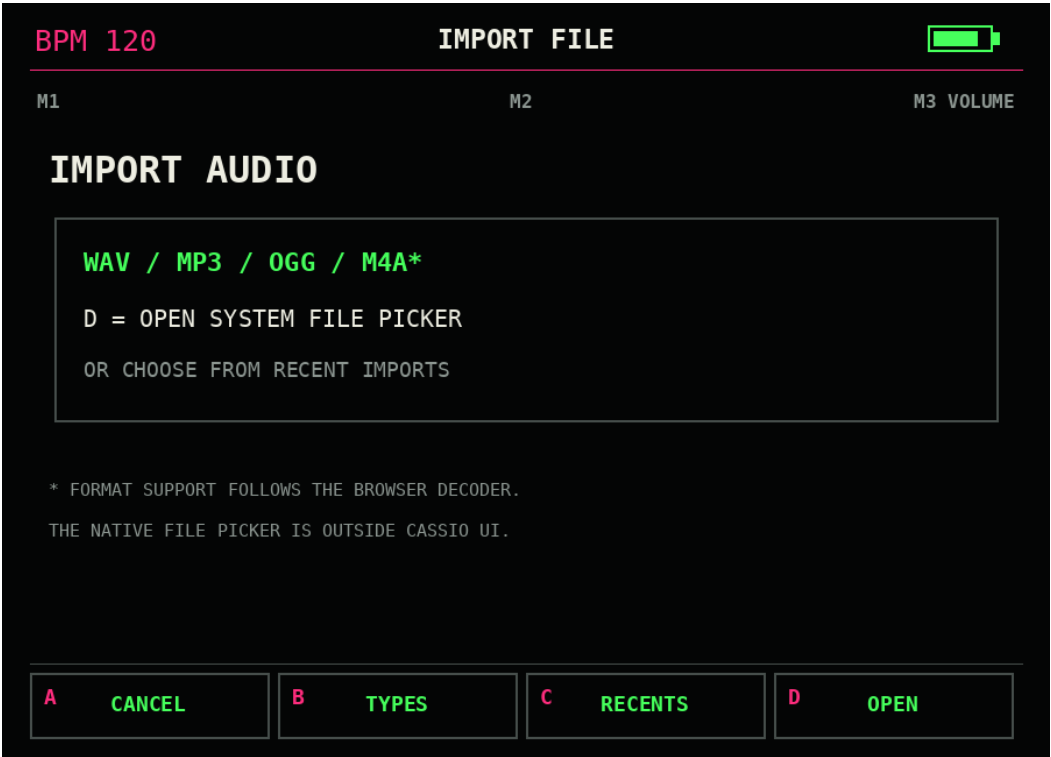


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Sampler Home -> A MIC.
HARDWARE	REC starts recording after the configured count-in; STOP ends the take; PLAY auditions the last take. M1 = digital input gain. M2 = record threshold when threshold mode is enabled. B toggles count-in. C toggles monitoring. D saves/continues after a take.
WHAT IT DOES	The mic signal is captured into an audio buffer. Monitoring is OFF by default to prevent speaker feedback. Input gain changes recorded level; it does not change microphone hardware sensitivity.
SAVE / STATE	The take remains temporary until Save Sample. A staged take can be replaced by pressing REC again.
NOTES	On first use, the browser may request microphone permission. The onboard mic is the default/only V1 input source.

Import File

Brings an audio file from the host file system into the CASSIO user library.

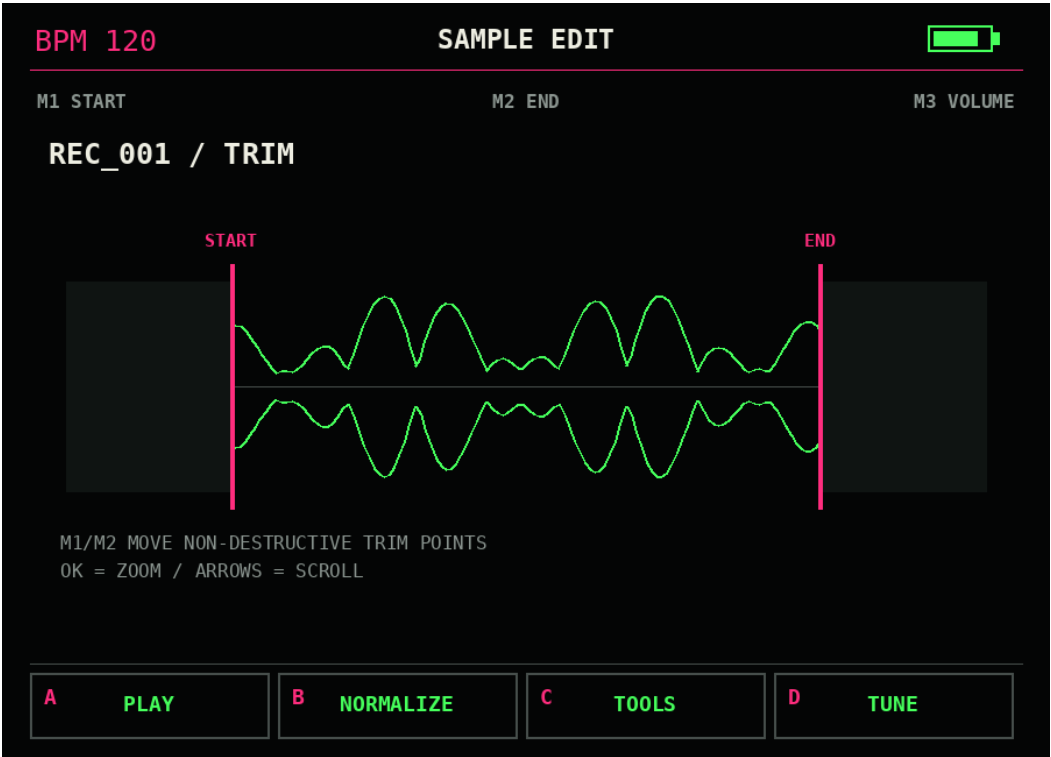


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Sampler Home -> B FILE.
HARDWARE	D = open system file picker. A = cancel. B displays supported formats. C can show recent imports. After the native picker returns a file, CASSIO decodes it and opens Sample Edit.
WHAT IT DOES	Importing does not alter the original file. CASSIO makes its own local working copy/reference and previews it through the same sound engine.
SAVE / STATE	The imported source is staged until Save Sample, then persisted in local app storage.
NOTES	Guaranteed V1 targets should be WAV and MP3. OGG and M4A/AAC can be accepted where the browser decoder supports them. The native file-picker UI itself is provided by the operating system/browser, not CASSIO.

Sample Edit - Trim

Defines the playable region of a recorded/imported source without destroying the original.

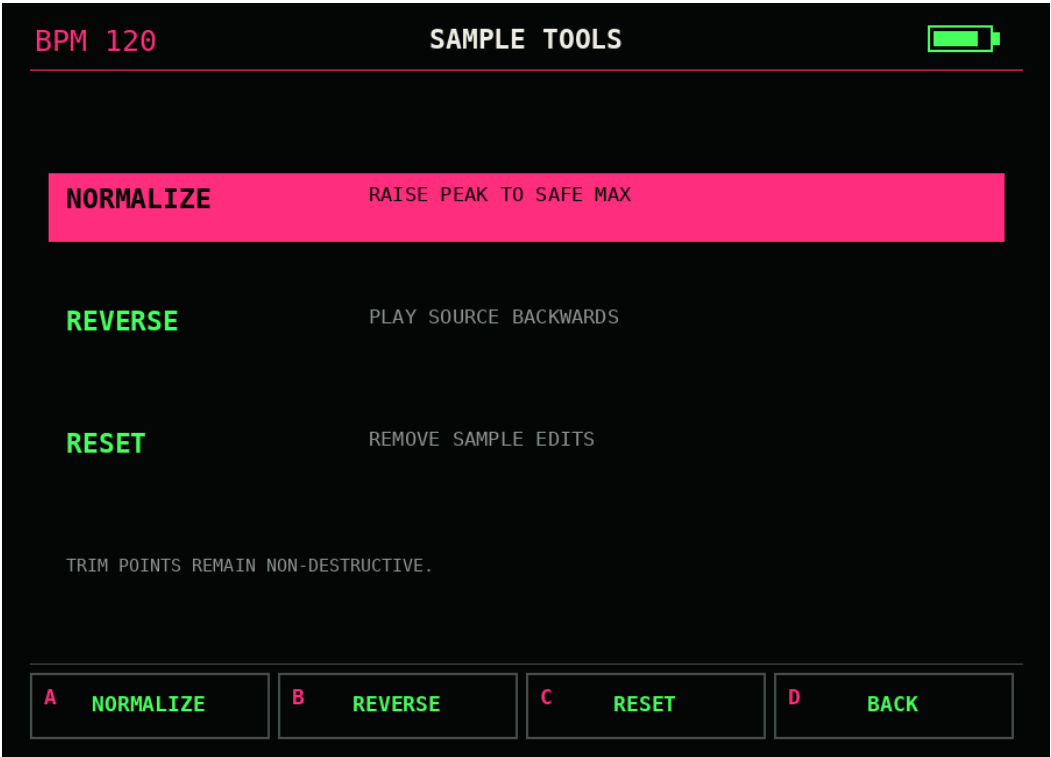


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	After mic recording/import, or open a user sample -> Edit -> Trim.
HARDWARE	M1 moves sample Start. M2 moves sample End. LEFT/RIGHT scroll the waveform; UP/DOWN zoom. A plays the selected region. B normalizes. C opens Sample Tools. D goes to Tune.
WHAT IT DOES	Only the selected region is heard when the sample is triggered. Normalization increases or decreases gain so the peak reaches a safe target without changing tone.
SAVE / STATE	Trim points and normalization metadata are saved with the user sample. Source audio remains intact.
NOTES	Keep Start before End; enforce a minimum playable region to prevent invalid edits.

Sample Tools

Utility operations for a sample source.

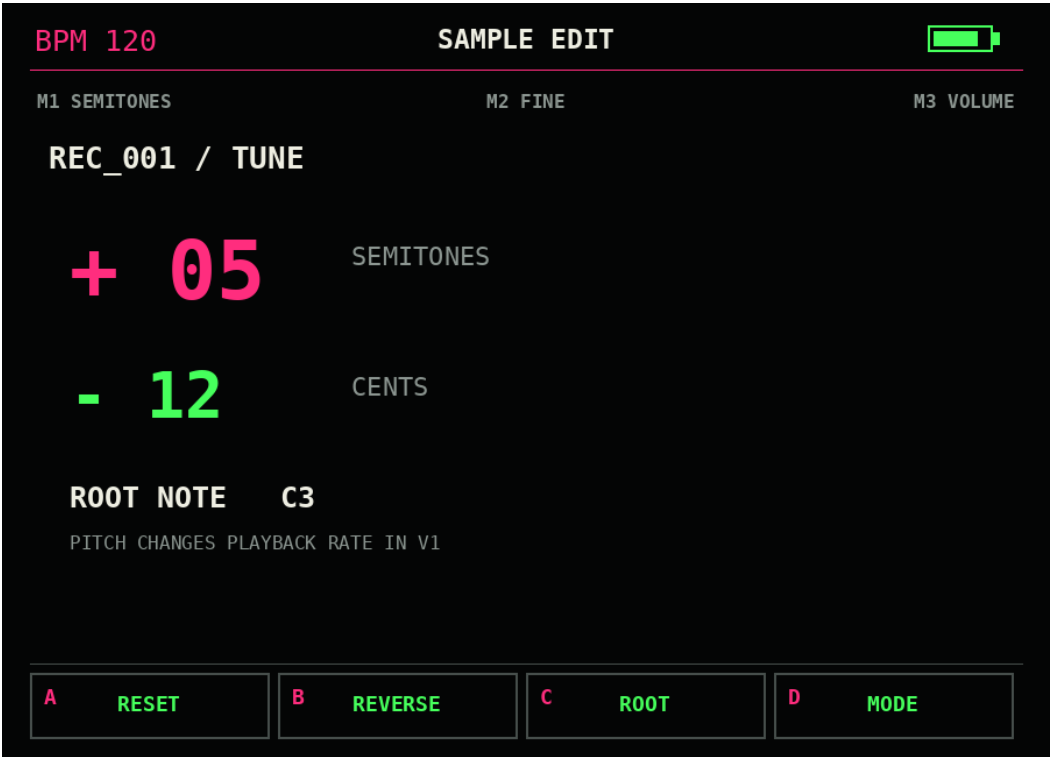


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Sample Edit - Trim -> C TOOLS.
HARDWARE	A normalize, B reverse, C reset sample edits, D back. UP/DOWN can highlight the same actions and OK confirms.
WHAT IT DOES	Normalize changes level only. Reverse plays the selected source region backward. Reset returns trim/reverse/normalization state to the original imported/recorded source.
SAVE / STATE	Tools remain non-destructive metadata until the sample is exported/rendered. Reset is therefore safe.
NOTES	Destructive deletion is never placed on this screen; sample deletion lives in the library/storage workflow and requires confirmation.

Sample Edit - Tune

Changes pitch mapping and the root note of a user sample.



HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Sample Edit -> Tune page.
HARDWARE	M1 = coarse semitone tuning. M2 = fine cents. C/ROOT lets arrows choose the root note. B/REVERSE can toggle reverse if desired. A/RESET restores tuning. D/MODE opens playback behavior.
WHAT IT DOES	Coarse/fine tuning changes playback rate in V1, so pitch and playback duration change together. Root note tells the keyboard which physical note plays the sample at its original saved pitch.
SAVE / STATE	Tuning/root are stored with the user sample preset.
NOTES	Pitch-preserving time stretch is intentionally not required for V1; it can be added later as software only.

Sample Edit - Playback

Defines how a sample responds to a key/pad trigger.



HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Sample Edit -> Mode/Playback page.
HARDWARE	A = One-shot. B = Gate. C = Loop. M1 = Attack smoothing. M2 = Release smoothing. D = EQ.
WHAT IT DOES	One-shot plays to the end after a trigger. Gate plays only while held. Loop repeats the selected region until released/stopped. Attack/Release smooth clicks at boundaries.
SAVE / STATE	Playback mode and smoothing values save with the user sample.
NOTES	Pads commonly default to One-shot; keyboard-mapped sustained sounds may prefer Gate or Loop.

Sample Edit - EQ

Two-band tone shaping for recorded or imported samples before they are used as instruments.



HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Sample Edit -> EQ page.
HARDWARE	M1 = Bass, M2 = Treble, each -12 dB to +12 dB. A/B/C/D return to Trim/Tune/Mode/EQ tabs.
WHAT IT DOES	Bass changes low-end weight; Treble changes brightness/noise/air. EQ is applied during playback and is not baked into the source file unless exported.
SAVE / STATE	Values save with the sample preset.
NOTES	After saving, the resulting user sound can also use the generic Sound Edit pages for filter/envelope/FX.

Save Sample

Commits the staged recording/import as a reusable user sound.

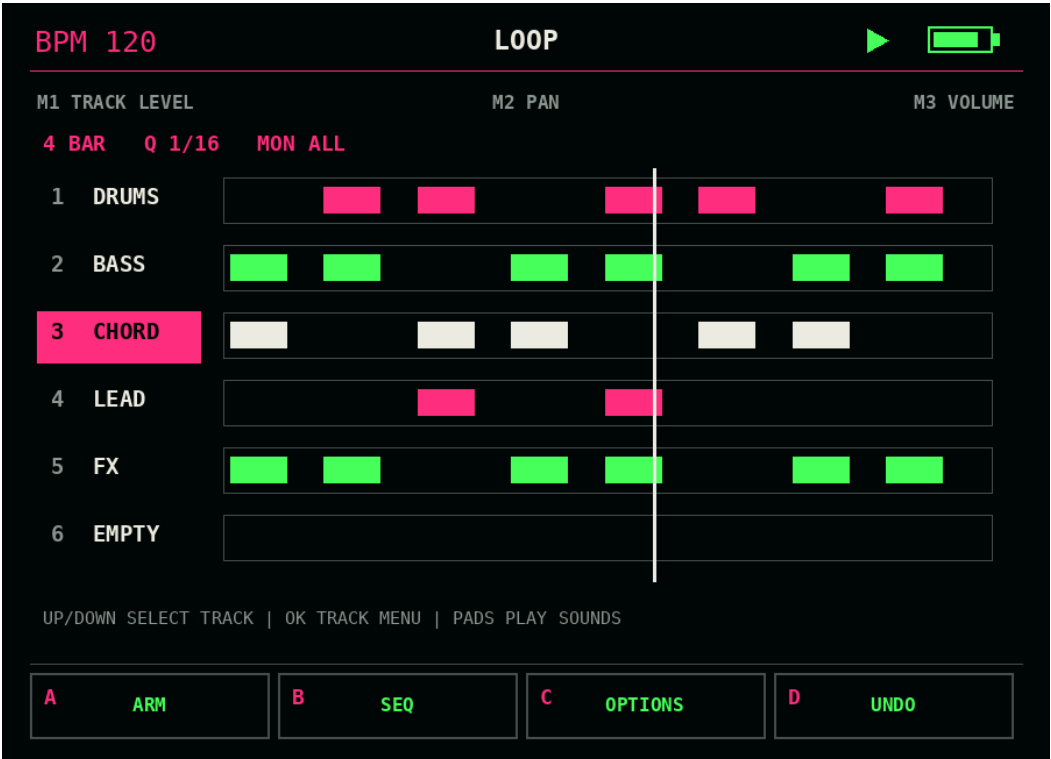


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Press BACK/Save from the sample editor after a usable take exists.
HARDWARE	A cancel, B name, C assign, D save. Naming uses Text Entry. Assign can route the new sound to the keyboard or a pad before leaving.
WHAT IT DOES	Saving itself does not process the audio beyond the current non-destructive edit chain. The newly saved sound is immediately playable.
SAVE / STATE	Creates a user-library item referencing the local source plus trim/tune/mode/EQ metadata.
NOTES	The original source remains available until storage cleanup determines it is unreferenced.

LOOP - Track View

Primary multitrack looper. Six tracks are stacked vertically with a common playhead.



HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	MENU -> LOOP, or REC from a performance screen when a loop track is selected.
HARDWARE	UP/DOWN selects track; OK opens Track Menu. M1 = track level. M2 = pan. A = ARM, B = SEQ, C = OPTIONS, D = UNDO. Keyboard and pads remain playable. Transport controls record/play/stop.
WHAT IT DOES	REC from stopped records after count-in; REC while playing enters overdub. Mute/Solo and record mode live in OK -> Track Menu. Backing-track monitoring follows Loop Options.
SAVE / STATE	Loop audio, track level/pan/mute state and selected track save with the project.
NOTES	Pads remain performance inputs here; track selection stays on the D-pad so pad-based drum performances can be recorded.

Loop Track Menu

Per-track record and monitoring options.

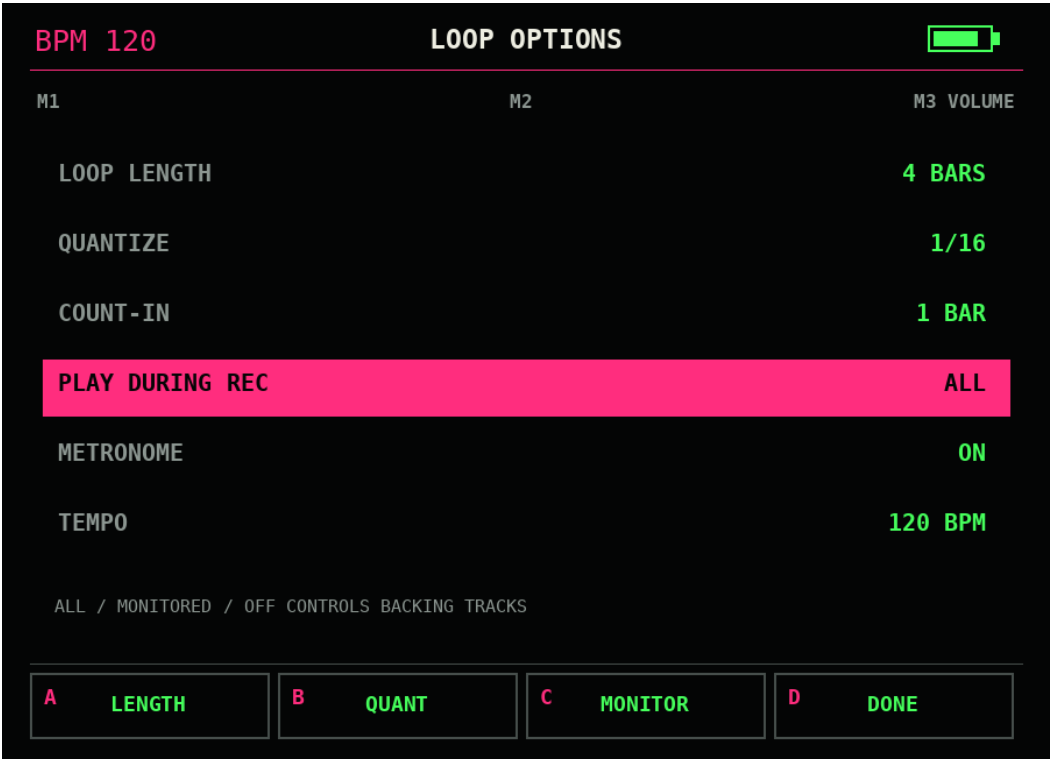


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	LOOP -> select a track -> OK.
HARDWARE	UP/DOWN selects Record Mode, Monitor During Rec, Mute, Solo, Level or Pan; LEFT/RIGHT or OK changes the selected row. A shortcuts Record Mode, B toggles Monitor, C clears after confirmation, D returns. M1/M2 continue level/pan.
WHAT IT DOES	Overdub adds new performance to existing audio; Replace records a fresh pass. Mute/Solo affect playback. Monitor controls whether the track is heard when Loop Options uses MONITORED recording playback.
SAVE / STATE	Track options save in the project. Clear removes recorded material but not source sounds.
NOTES	Clear is destructive and therefore uses the common confirmation pattern.

Loop Options

Global loop timing, quantization, count-in and backing-track behavior.

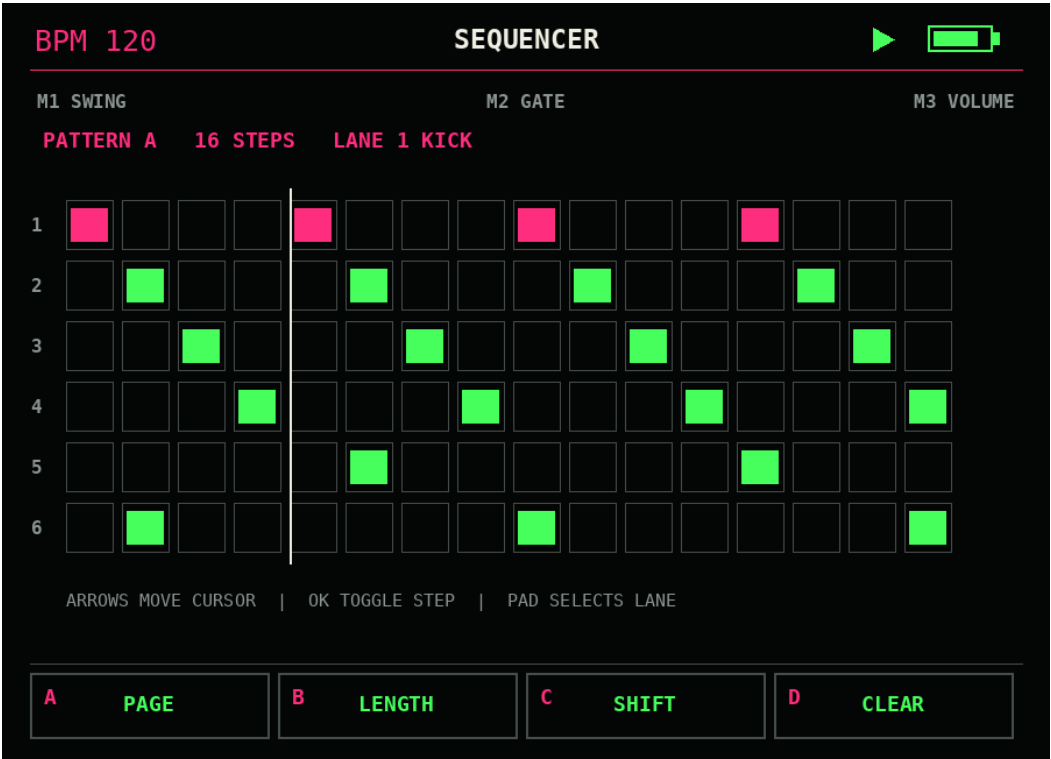


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	LOOP -> C OPTIONS.
HARDWARE	UP/DOWN selects Length, Quantize, Count-in, Play During Rec, Metronome or Tempo. LEFT/RIGHT changes values. A-D jump to the most common option groups.
WHAT IT DOES	Loop Length defines the repeating region. Quantize aligns recorded events to the grid. Count-in delays recording. Play During Rec = ALL, MONITORED or OFF controls which existing tracks are heard while recording.
SAVE / STATE	Options save with the project.
NOTES	Recommended V1 loop lengths: 1, 2, 4 or 8 bars. Quantize: Off, 1/4, 1/8, 1/16.

Step Sequencer

Programs repeating step patterns without live recording.



HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	LOOP -> B SEQ.
HARDWARE	Pads 1-6 select a sequencer lane on this screen only. Arrows move the step cursor. OK toggles a step. M1 = swing; M2 = gate. A page, B length, C shift/nudge, D clear lane after confirmation. PLAY/STOP auditions the pattern.
WHAT IT DOES	Active steps trigger the sound assigned to their lane. Swing delays alternating subdivisions; Gate changes note/sample duration when supported.
SAVE / STATE	Pattern, lane assignments, length, swing/gate and step states save in the project.
NOTES	The base grid is 16 steps; longer patterns use pages rather than a wider physical display.

Step Edit

Fine control for one sequencer step.

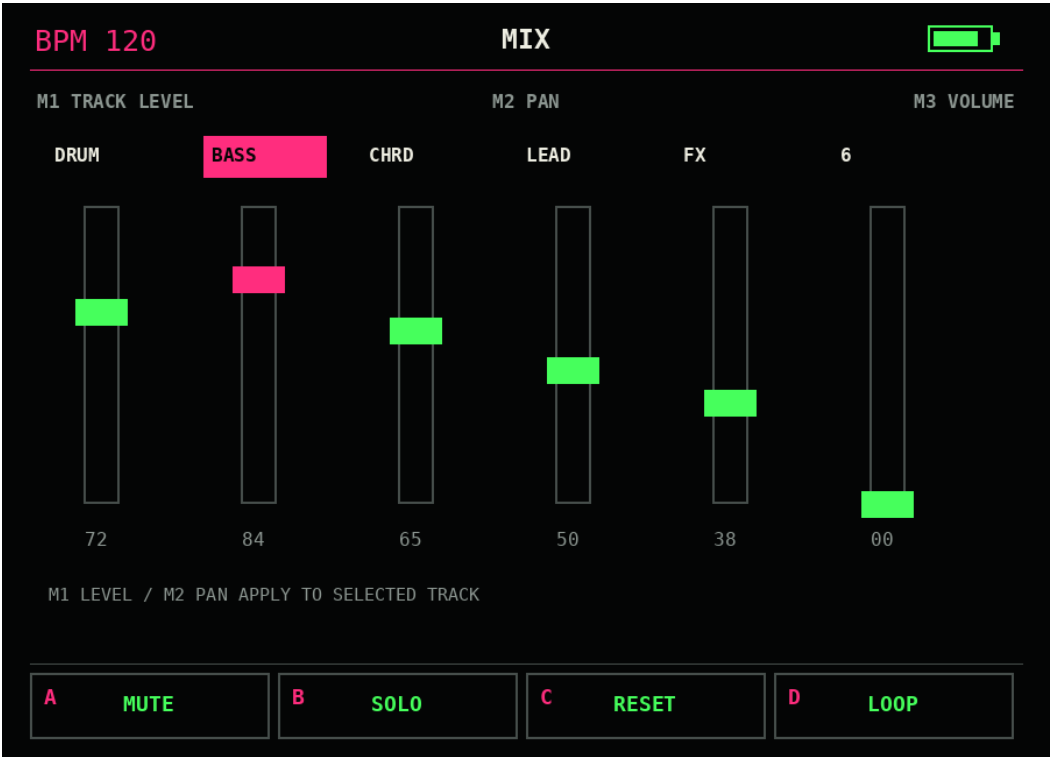


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Sequencer -> highlight a step -> OK/hold OK depending implementation.
HARDWARE	M1 = step velocity/value. M2 = gate. A off, B accent, C tie, D done. LEFT/RIGHT can set micro-shift where enabled.
WHAT IT DOES	Velocity changes trigger level/intensity for the step. Accent is a quick high-velocity state. Tie holds a note across the next step. Micro-shift nudges timing without moving the visible grid location.
SAVE / STATE	Step values save in the pattern.
NOTES	If the physical keys/pads are not velocity sensitive, sequencer velocity still works because it is a software parameter.

Mixer

Balances the six loop tracks without changing their source recordings.

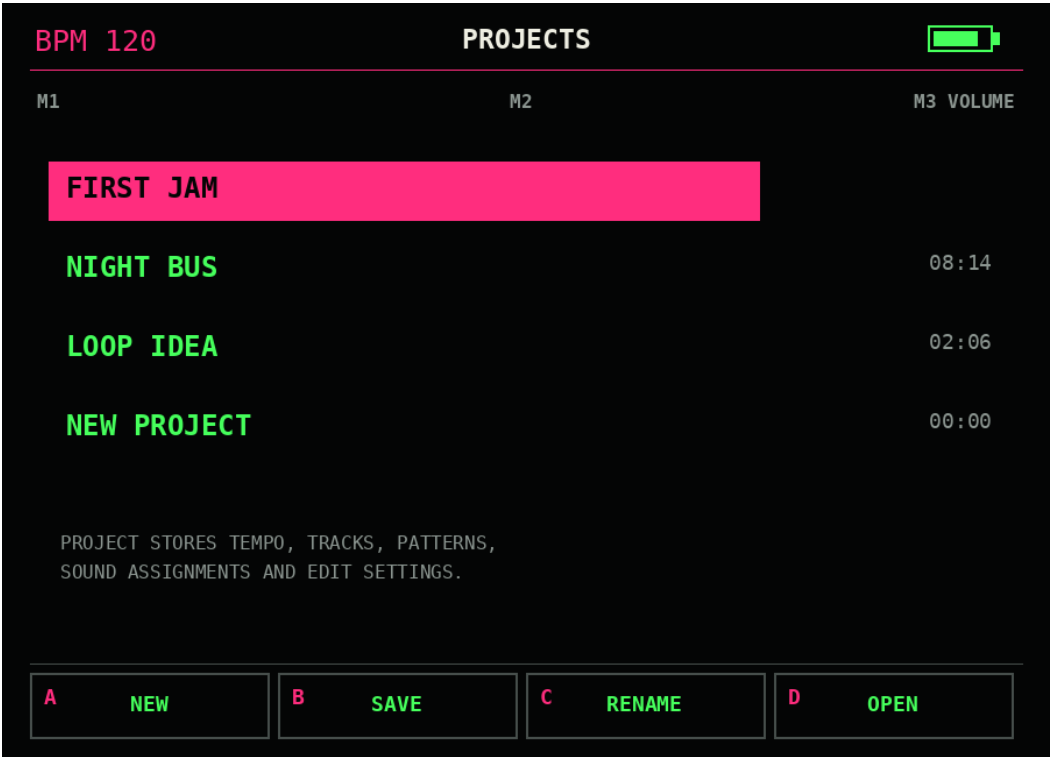


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	MENU -> MIX, or LOOP -> Mix shortcut.
HARDWARE	LEFT/RIGHT or UP/DOWN selects track. M1 = track level. M2 = pan. A mute, B solo, C reset selected track mix, D return to Loop.
WHAT IT DOES	Level changes loudness of the track in the mix; pan moves it left/right. Mute silences a track; Solo temporarily isolates it. None of these alter the recorded audio.
SAVE / STATE	Mixer state saves with the project.
NOTES	M3 still controls final master volume after the track mix.

Projects

Lists named CASSIO projects and opens project-management actions.



HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	MENU -> PROJECT.
HARDWARE	UP/DOWN selects a project. A new, B save current, C rename selected, D open selected. OK can also open.
WHAT IT DOES	Opening a project replaces the active tempo, sounds, loop tracks, pattern and mix state. Current unsaved changes must be saved/discarded first.
SAVE / STATE	Projects are stored locally in persistent browser/app storage and reference user samples by ID.
NOTES	CASSIO should also maintain an automatic recovery snapshot so an unexpected shutdown does not erase the current session.

Save Project

Explicitly saves the current musical state or creates a copy.

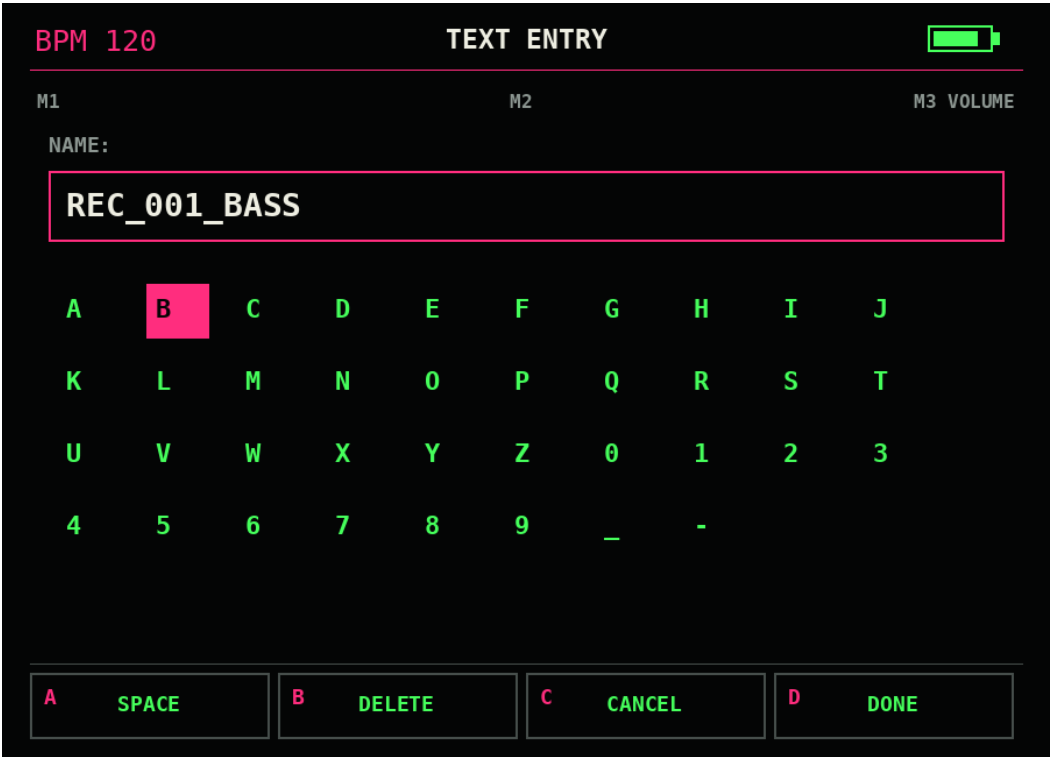


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	PROJECT -> B SAVE, or leaving a dirty project when prompted.
HARDWARE	A cancel, B name, C Save As, D Save. New projects use Text Entry for the name.
WHAT IT DOES	Saving does not interrupt audio beyond any brief storage operation; it only commits the current state.
SAVE / STATE	Stores tempo/key metadata, sound assignments, user-sound edit values, loop tracks, sequencer patterns and mixer settings.
NOTES	Large user audio sources are stored once and referenced, rather than duplicated into every project.

Text Entry

Reusable character-grid screen for naming projects and user sounds without adding a hardware keyboard.

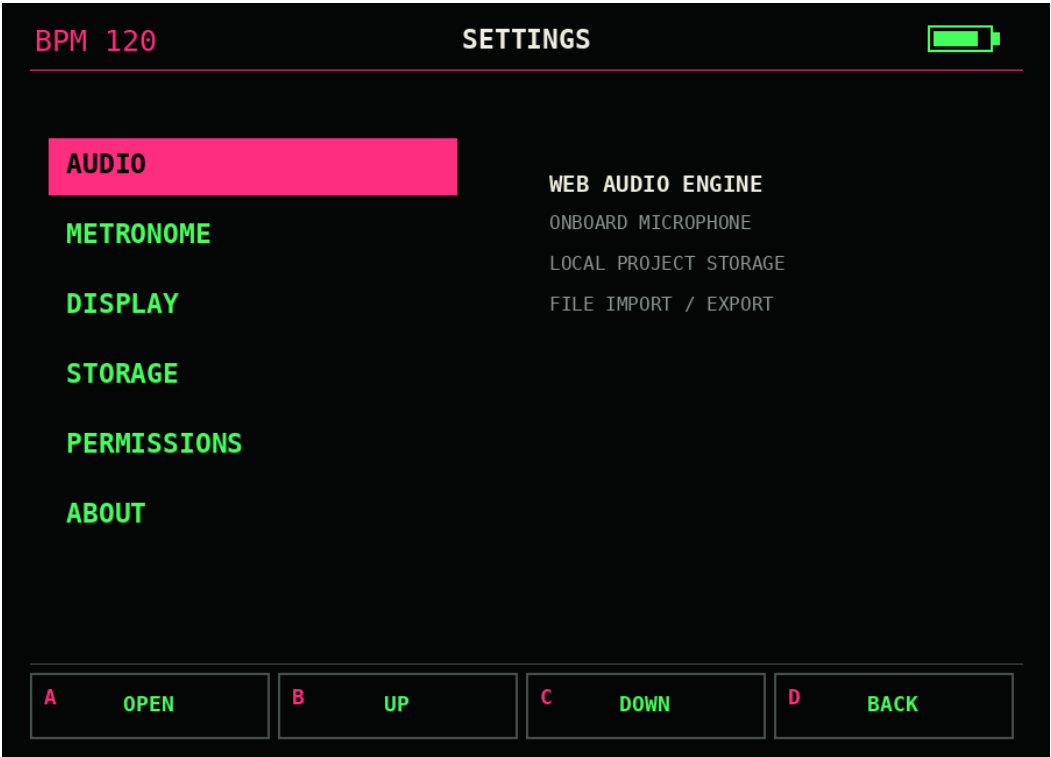


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Any Name/Rename action.
HARDWARE	Arrows move cursor. OK inserts character. A space, B delete, C cancel, D done. Holding LEFT/RIGHT may accelerate cursor movement.
WHAT IT DOES	No audio change. Playback can remain paused or continue depending on the parent screen.
SAVE / STATE	The completed name is returned to the parent Save/Rename screen.
NOTES	Keep allowed names compact and filesystem-safe: letters, digits, spaces, dash and underscore are sufficient for V1.

Settings Home

Entry point for system behavior that is not part of a musical project.

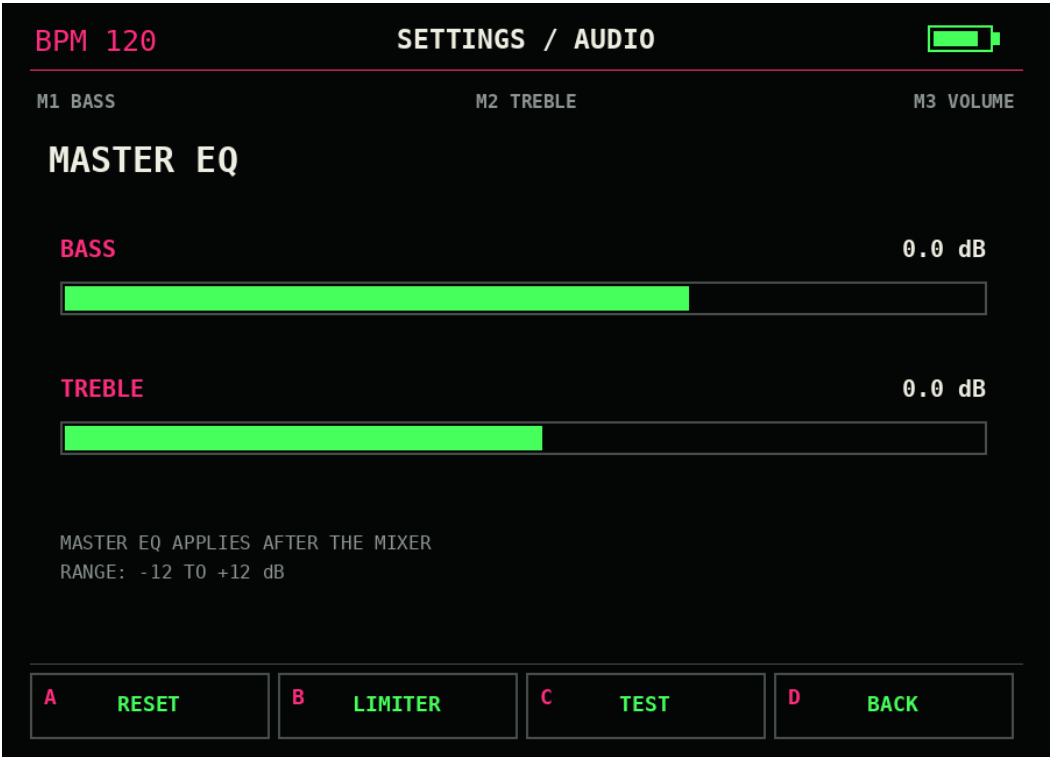


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	MENU -> SETTINGS.
HARDWARE	UP/DOWN selects Audio, Metronome, Display, Storage, Permissions or About. OK/A opens. BACK returns.
WHAT IT DOES	Merely opening Settings does not alter sound. Audio/Metronome pages can make live changes as the user adjusts them.
SAVE / STATE	System preferences persist independently of projects.
NOTES	This menu is also where future software-only settings can be added without changing hardware.

Settings - Master Audio

Global output EQ and safety processing after all tracks/sounds have been mixed.



HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	SETTINGS -> AUDIO.
HARDWARE	M1 = master Bass, M2 = master Treble. A reset EQ. B limiter toggle. C test tone/preview. D back. M3 remains master volume.
WHAT IT DOES	Bass/Treble alter the whole device output. Limiter catches peaks caused by stacked tracks/EQ boosts and reduces clipping.
SAVE / STATE	Master EQ and limiter preference persist globally, not per project.
NOTES	Per-sound EQ and master EQ are intentionally separate: one shapes a sound; the other compensates for speakers/headphones or overall taste.

Settings - Metronome

Configures the click and recording count-in.



HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	SETTINGS -> METRONOME.
HARDWARE	M1 click level. M2 count-in bars. A chooses click sound, B count-in mode, C level/reset, D back. Hold TAP toggles metronome from any screen.
WHAT IT DOES	Metronome adds a click to playback/recording monitoring only; it is never recorded into loop audio. Count-in plays before recording begins.
SAVE / STATE	Settings persist globally; a project stores BPM but not necessarily the preferred click sound.
NOTES	Beat 1 may use an accented click so bar boundaries are obvious.

Settings - Display

Controls the hardware display brightness and inactivity behavior.

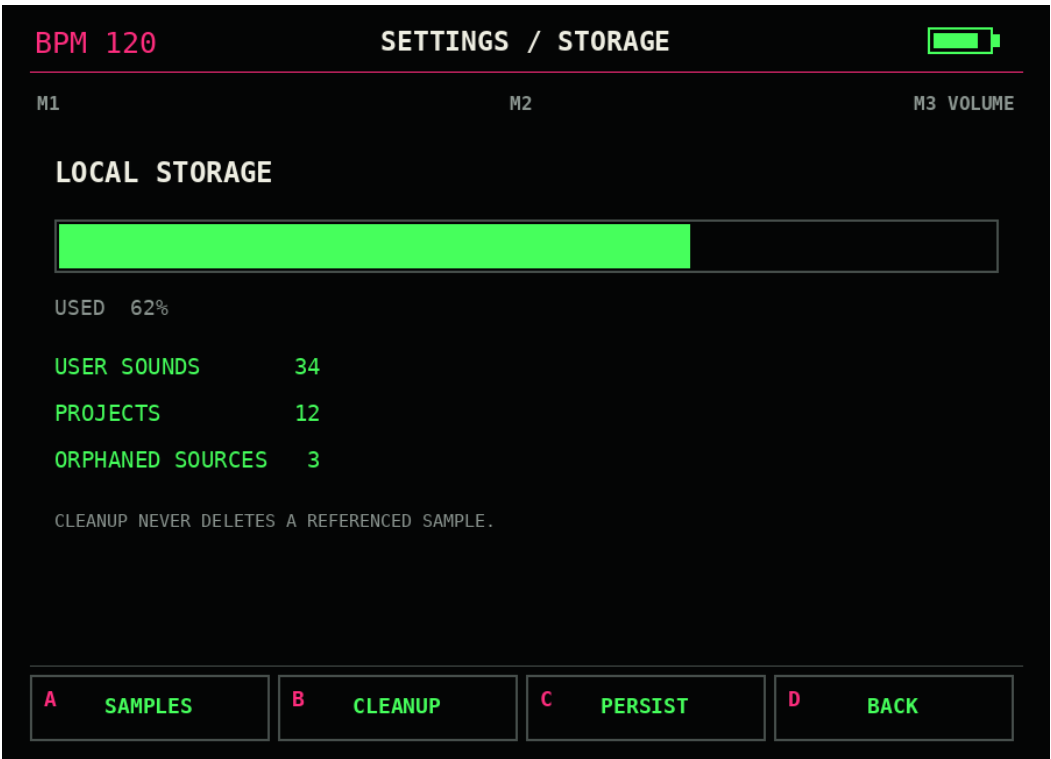


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	SETTINGS -> DISPLAY.
HARDWARE	M1 brightness. M2 dim level. UP/DOWN chooses auto-dim or screen-sleep time; LEFT/RIGHT changes values. D back.
WHAT IT DOES	No audio effect.
SAVE / STATE	Display preferences persist globally.
NOTES	Audio playback/recording must not stop merely because the screen dims. Screen sleep is a visual power-saving state only.

Settings - Storage

Shows local storage use and manages user media safely.

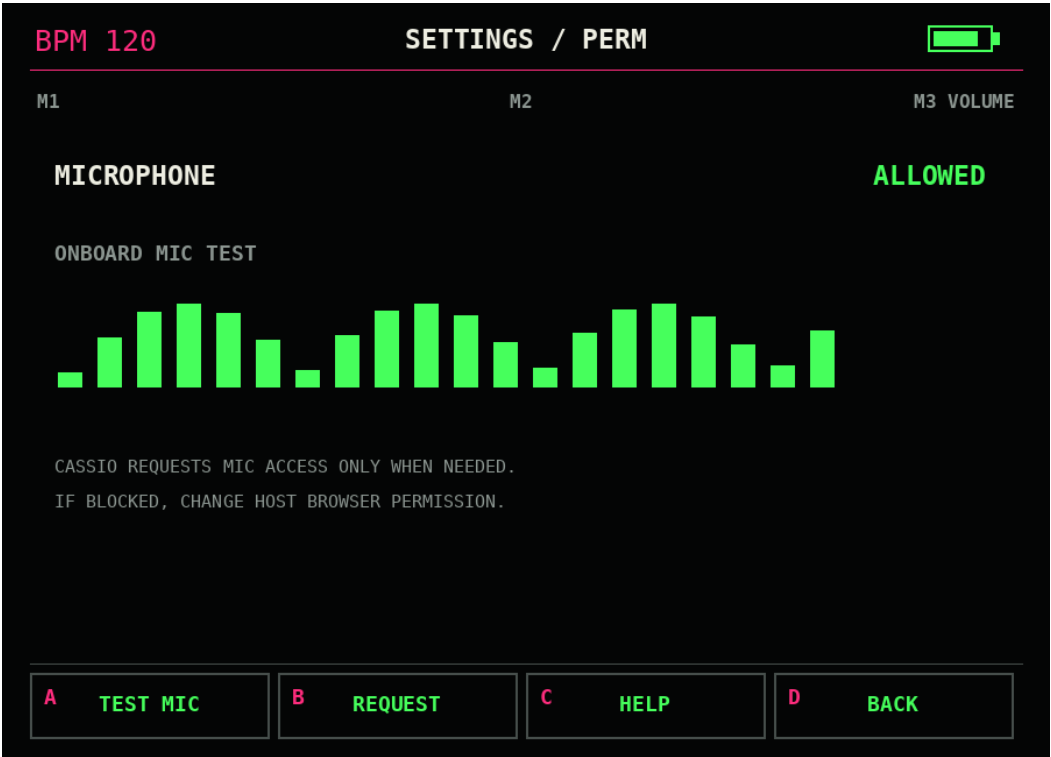


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	SETTINGS -> STORAGE.
HARDWARE	A jumps to user samples (same USER library screen). B cleanup. C requests/presents persistent-storage status. D back. Arrows navigate rows.
WHAT IT DOES	No audio effect unless the user deletes an unreferenced sample.
SAVE / STATE	Cleanup may remove orphaned source files only after verifying that no project/user sound references them. Persistent storage requests the browser/device to reduce automatic eviction.
NOTES	Never delete a referenced sample silently. Deletion actions use a confirmation dialog.

Settings - Permissions

Checks and repairs onboard microphone access.



HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	SETTINGS -> PERMISSIONS.
HARDWARE	A tests mic and shows level meter. B requests permission if available. C shows help text. D back.
WHAT IT DOES	Mic Test meters input; monitoring remains off, so testing should not feed the mic directly to speakers.
SAVE / STATE	Permission state is controlled by the host browser/OS; CASSIO stores only its own preference/status cache.
NOTES	On a dedicated device, microphone access can ideally be pre-authorized so the user never sees a browser permission prompt.

About

Read-only software/build information.

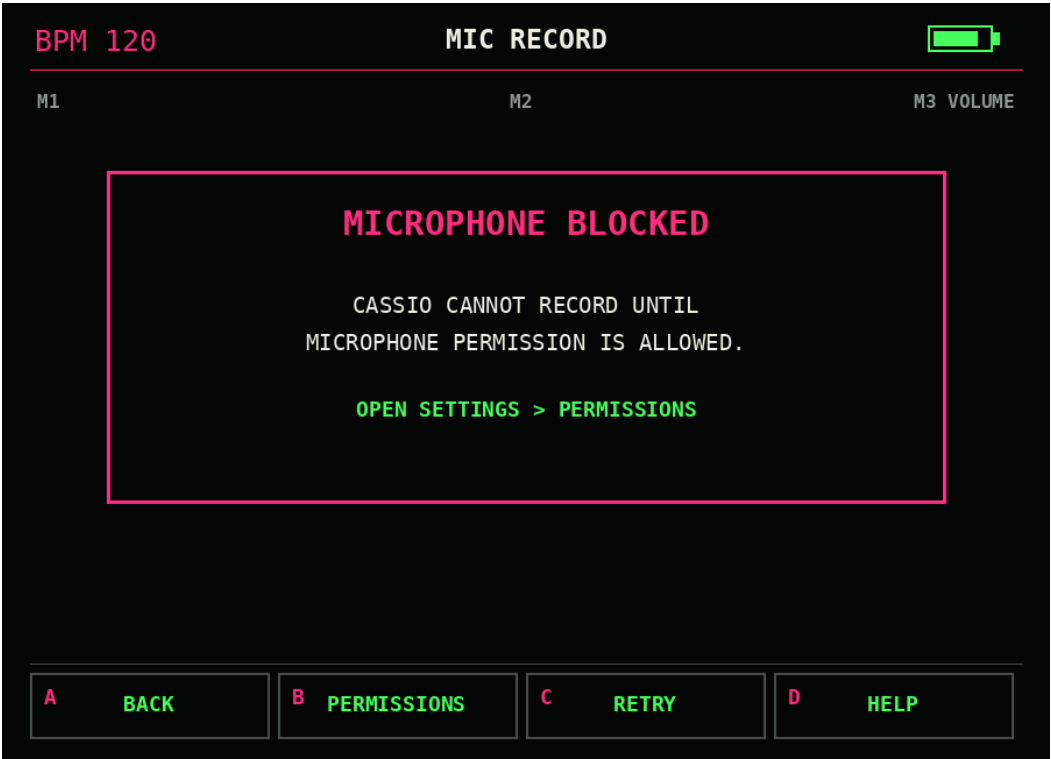


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	SETTINGS -> ABOUT.
HARDWARE	D/Back returns. Other soft keys may be blank in a production build unless diagnostics are deliberately exposed.
WHAT IT DOES	No audio effect.
SAVE / STATE	No musical state changes.
NOTES	The production screen should identify the device as a pocket synth/sampler and list Web Audio, onboard microphone, file import and local storage capabilities.

Microphone Blocked

Error state shown when the sampler cannot access the onboard microphone.

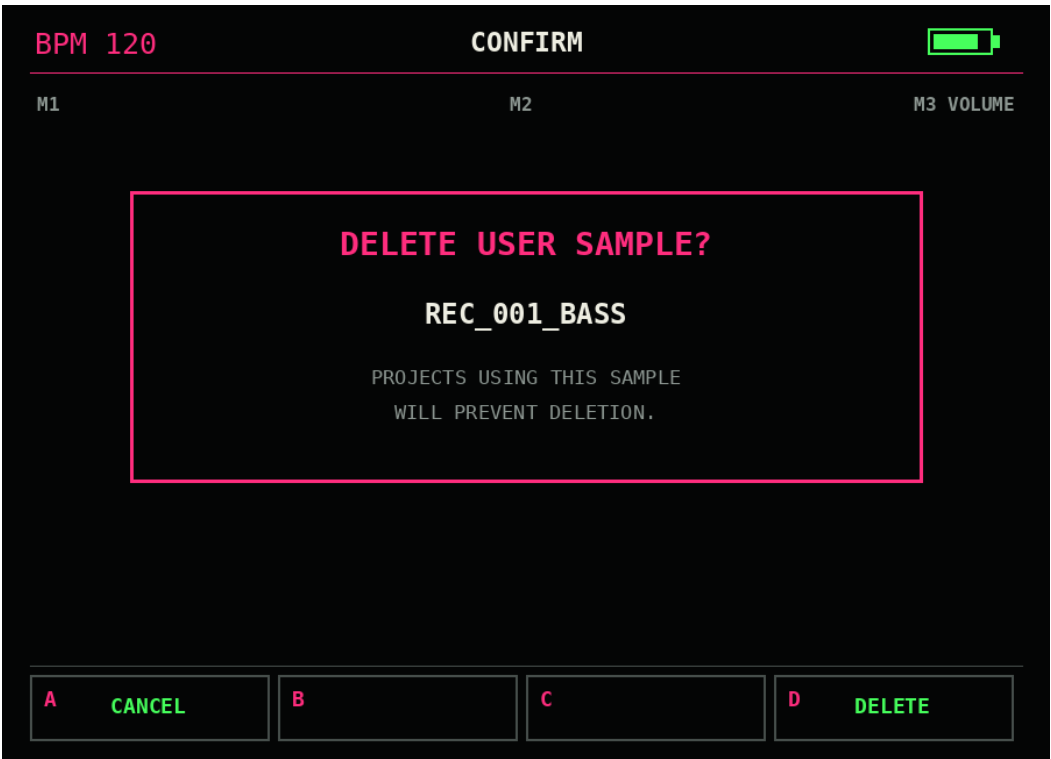


HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Appears after Mic Record or Mic Test when browser permission is denied/unavailable.
HARDWARE	A back, B open Permissions page, C retry, D help. REC remains disabled until permission is available.
WHAT IT DOES	No mic audio is captured. Existing instrument/loop playback is unaffected.
SAVE / STATE	No recording is created.
NOTES	The error must explain the next action instead of failing silently.

Destructive Confirmation

Reusable confirmation pattern for deleting a user sample, clearing a loop track, clearing a sequencer lane or storage cleanup.



HARDWARE DISPLAY REFERENCE - software mockup for the V1 implementation

GET THERE	Appears only after a destructive action.
HARDWARE	A cancel. D confirm destructive action. The middle soft keys remain blank unless the specific action needs another choice.
WHAT IT DOES	Sound changes only after confirmation. Cancelling returns exactly to the previous state.
SAVE / STATE	Deletion/clear is committed only after D confirm. Referenced samples cannot be deleted; CASSIO should identify the blocking project/sound.
NOTES	Using one consistent destructive-dialog pattern reduces accidental data loss.

V1 factory sound library

This is the starter library specification for the shipped V1 experience. Factory sounds are read-only and can be copied into the User library after editing. The exact underlying audio/synthesis assets still need to be authored, recorded or licensed, but the categories, names and macro intent can be implemented now.

SYNTH

SOUND	M1	M2
Glass Poly	Brightness	Space
Neon Lead	Cutoff	Drive
Bit Pluck	Tone	Decay
Tape Pad	Color	Space
Pulse Stack	Width	Motion
Soft Chime	Tone	Release
Hollow Bell	Color	Decay
Rubber Mono	Cutoff	Bite

BASS

SOUND	M1	M2
Round Bass	Tone	Decay
Sub Mono	Harmonics	Release
Rubber Bass	Cutoff	Bite
FM Bass	Color	Snap
Dust Bass	Tone	Drive
Pluck Bass	Brightness	Decay

KEYS

SOUND	M1	M2
Warm EP	Tone	Tremolo
Toy Piano	Color	Decay
Tape Piano	Age	Space
Soft Organ	Tone	Rotary
Bell Keys	Brightness	Release
Lo-Fi Keys	Filter	Wow

DRUMS

SOUND	M1	M2
Kick 01	Tone	Decay
Kick 02	Punch	Decay
Snare 01	Tone	Snap
Snare 02	Body	Noise
Clap	Tone	Width
Closed Hat	Tone	Decay

Open Hat	Tone	Decay
Low Tom	Tone	Decay
High Tom	Tone	Decay
Rim	Tone	Room
Shaker	Tone	Decay
Perc Hit	Tone	Space

TEXTURE

SOUND	M1	M2
Cassette Air	Color	Motion
Vinyl Dust	Tone	Density
Digital Rain	Color	Motion
Breath Pad	Tone	Space

FX

SOUND	M1	M2
Rise	Tone	Length
Fall	Tone	Length
Impact	Body	Space
Glitch Hit	Color	Repeat

Factory, User, Recordings and Favorites

FACTORY	Bundled read-only sounds. Editing always creates a User copy. Factory assets cannot be deleted.
USER	Saved user sounds created from edited factory sounds, microphone recordings or imported audio.
RECORDED / IMPORTED	Filters inside the User library for sounds created from the onboard microphone or imported files; not separate copies.
FAVORITES	References to sounds in Factory or User. Removing a favorite never deletes the sound.
SOURCE AUDIO	A recorded/imported source is stored once. Multiple user sounds may reference it with different trim/tune/EQ/edit settings.
DELETE	Allowed only when no project or user sound references the item/source. Destructive confirmation is mandatory.
EXPORT	User sample sources may be exported as WAV through the browser's normal download/save flow. Rendered Copy applies current edits; Source Export returns the original.
STORAGE	Use persistent browser/app storage when available. Show usage and warn before storage pressure becomes a recording failure.

What the sound controls actually do

TONE / CUTOFF	Reduces or reveals upper harmonics. Lower values sound darker; higher values sound brighter.
RESONANCE	Boosts a narrow band around the filter cutoff. High values make the filter more pronounced/ringing.
ATTACK	Time from note trigger to full level. Short = immediate; long = fade-in.
RELEASE	Time for sound to fade after key release. Short = tight; long = lingering.
BASS EQ	Low-shelf gain, recommended center around 120 Hz, -12 to +12 dB.
TREBLE EQ	High-shelf gain, recommended center around 6 kHz, -12 to +12 dB.
REVERB	Adds reflected/room-like tail. Amount sets wet level.
DELAY	Adds repeats/echo. V1 exposes amount; timing defaults may follow tempo/preset.
SAMPLE START/END	Defines the region read from the source buffer.
NORMALIZE	Applies gain so the loudest peak reaches a safe target. It does not compress dynamics.
REVERSE	Reads the selected sample region backward.
COARSE/FINE TUNE	Changes sample playback rate. In V1 this changes pitch and duration together.
ROOT NOTE	Defines the keyboard note that plays a sample at its saved/base pitch.
ONE-SHOT	A trigger starts the sample and it continues to the end.
GATE	The sample plays only while the triggering key/pad remains held.
LOOP	The selected sample region repeats until release/stop.
TRACK LEVEL	Non-destructive volume for a loop track.
PAN	Moves a track between left and right output channels.
SWING	Delays alternating sequencer subdivisions to create a less rigid feel.
GATE (SEQ)	Sets sequenced note duration relative to the step.

Recording, overdub and monitoring rules

These rules are important because they define what the user hears while creating a loop.

NEW RECORD	REC while stopped starts the selected track after count-in. The track becomes the timing reference if the project has no existing loop.
OVERDUB	REC while playback is running adds a new pass to the selected track. Press REC again to stop adding while playback continues.
REPLACE	If the selected track's record mode is Replace, the next record pass replaces that track after confirmation/count-in rather than layering.
PLAY DURING REC: ALL	All unmuted existing tracks are audible while recording.
PLAY DURING REC: MONITORED	Only tracks whose Track Menu Monitor flag is ON are audible while recording.
PLAY DURING REC: OFF	Backing tracks are silent while recording; metronome may still play.
UNDO	Removes only the most recent record/overdub operation. It should not undo unrelated sound-library edits.
METRONOME	Never becomes part of the recorded loop audio.
MIC SAMPLE MONITOR	Off by default to prevent feedback. The user may turn it on deliberately for headphones.
PAD/KEY MONITOR	The user always hears the sound they are currently playing while recording unless a modal dialog blocks performance input.

What is saved where

GLOBAL SETTINGS	Master EQ, limiter preference, metronome click preference, display settings, storage permission/status and mic permission status cache.
PROJECT	BPM, key metadata, keyboard sound, pad assignments, loop tracks, loop options, sequencer patterns, mixer state, octave/hold/voice state where appropriate.
USER SOUND	Source reference, trim region, tuning/root, playback mode, envelope, EQ, filter/shape, FX and name/favorite metadata.
SOURCE AUDIO	Recorded mic take or imported file data. Stored once and referenced by user sounds/projects.
RECOVERY	A rolling autosave snapshot of the current session so a crash/power loss can reopen close to the last state.
FACTORY LIBRARY	Read-only application assets. Never modified by user edits.

Recommended browser implementation: use IndexedDB or Origin Private File System for local media/state, request persistent storage where supported, and use the browser file picker/download flow for imports/exports. The user manual should not expose those implementation names unless needed for troubleshooting.

V1 boundaries that keep the hardware stable

NO MIDI CONTROL	The web-app instrument does not depend on browser access to external MIDI hardware. The front panel is self-contained.
NO USB WORKFLOW	Charging/storage connector hardware may exist, but CASSIO V1 does not present a USB device-management screen or rely on USB for sound import.
FILE IMPORT	Uses the host browser/OS file picker. Once a file is returned to the web app, all editing is controlled from CASSIO hardware.
PITCH/TIME	V1 sampler tuning changes playback rate, so pitch and duration move together. Pitch-preserving time stretch can be added later digitally.
DEEP SYNTHESIS	V1 exposes musical macro controls rather than a full oscillator/modulation matrix. Deeper pages can be added later without changing hardware.
MORE TRACKS	The physical six-pad layout does not prevent future projects from having more digital tracks, but V1 fixes the loop/mixer to six for clarity.
TOUCH	No core V1 workflow requires touch. Native browser file/permission dialogs are the only host-controlled UI exceptions.

PHYSICAL HARDWARE CAN NOW BE FROZEN

Every V1 task in this manual is reachable using the current display, M1/M2/M3, A-D soft keys, six pads, BACK/MENU, the navigation cluster, transport buttons and mini keyboard. Future work can therefore concentrate on the digital interface, sound assets and audio engine without reopening the front-panel layout.